

2.4 Refinements to improve clarity and efficiency

وضاحت بہتر کرنے کے لیے معنی خیز نام، ماڈیول، درست indentation اور مختصر تبصرے استعمال کریں۔

CS XII

Chapter 2: Computational Thinking & Algorithms

Lecture 13 of 24

2.4 Refinements to improve clarity and efficiency

LEARN IT

Improving an algorithm means reviewing it step by step: simplifying logic, removing redundant work, and picking smart data structures. A clear algorithm is easier to follow; an efficient one wastes fewer steps.

2.4.1 Refinements for clarity

1. **Use meaningful names.** Replace vague variables with descriptive ones. Bad: `a = b / c`. Good: `average = totalSum / totalItems`.
2. **Break into modules.** Divide a massive block into functions, for example separating `CalculateSum()` from the printing logic.
3. **Improve structure and indentation.** Packed loops and IF statements confuse the reader. Spacing reduces mental effort.
4. **Add comments.** Keep them short. Explain **why** something is done, not just what is happening, especially for tricky logic.

2.4.2 Refinements for efficiency

1. **Reduce unnecessary steps.** In Bubble Sort, after the first pass the largest value has settled at the end. Ignore that sorted tail on the next pass by shrinking the inner loop, and you save comparisons.
2. **Use better algorithms.** Searching 1 million **sorted** records with Binary Search instead of Linear Search changes the time from seconds to milliseconds.
3. **Avoid re-computing values** (memoization / caching). If a value never changes, compute it once outside the loop, store it, and reuse it.
4. **Use appropriate data structures.** Direct access → array. Hierarchical data → tree. Do not force one structure everywhere.

WORKED EXAMPLE

Worked example 1 — shrinking Bubble Sort's inner loop ($n = 5$). A naive inner loop always runs $n-1 = 4$ times, for 4 passes: 16 comparisons. The refined inner loop runs 4, then 3, then 2, then 1: 10 comparisons.

```

n = 5
naive = 0
for i in range(n - 1):
    for j in range(n - 1):
        naive = naive + 1

refined = 0
for i in range(n - 1):
    for j in range(n - 1 - i):
        refined = refined + 1

print(naive, refined)

```

OUTPUT

```
16 10
```

Worked example 2 — do not re-compute a constant inside the loop.

```

A = 3
B = 4
N = 5

# BAD: C is recalculated every time
for i in range(1, N + 1):
    C = A + B
    print(C * i)

```

OUTPUT

```
7
14
21
28
35
```

```

A = 3
B = 4
N = 5
# GOOD: compute C once
C = A + B
for i in range(1, N + 1):
    print(C * i)

```

OUTPUT

```
7
14
21
28
35
```

Same output, fewer additions. That is caching: compute once, reuse N times.

REMEMBER IT

Clarity: **N-M-I-C** (Names, Modules, Indentation, Comments). Efficiency: **S-A-C-D** (Shrink steps, Algorithm choice, Cache, Data structure).

COMMON MISTAKE

Mistake: "optimising" by renaming variables and thinking that is an efficiency refinement.

Fix: meaningful names improve **clarity**. Efficiency refinements change the **amount of work** (fewer loop iterations, a better algorithm, no re-compute, a better structure).

اردو میں سمجھیں

وضاحت بہتر کرنے کے لیے معنی خیز نام، ماڈیول، درست indentation اور مختصر تبصرے استعمال کریں۔ کارکردگی کے لیے فضول قدم ہٹائیں (ببل سارٹ میں ترتیب شدہ دم نظر انداز کریں)، بہتر الگورتھم چنیں (لینئر کی جگہ بائنری سرچ)، لوپ کے اندر وہی حساب بار بار نہ کریں (Caching/Memoization)، اور مناسب ڈیٹا سٹرکچر رکھیں۔

CHECK YOURSELF

1. Rewrite $a = b / c$ with meaningful names for an average.
2. After Bubble Sort's first pass, what should the next pass ignore?
3. In the A+B example, where should $C = A + B$ be written?

Answers: 1. $average = totalSum / totalItems$. 2. The already-sorted largest value at the end (shrink the inner loop). 3. Once, outside the loop, then reuse C inside.

Next lecture: 2.5 Concept of Data Structure